

LAST NAME: _____

Practice Quiz 04D — MATH 2055 — Fall 2026

Practice for Quiz 04

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

Complements are taken in a fixed universal set. Name a definition or law at every step of part (a).

1. (a) Prove $(A \cap B) \cup (A \cap B^c) = A$ by a chain of equivalences beginning with membership in the left side.

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- (b) Let $\mathcal{F} = \{\{p\}, \{q\}\}$. Compute $\bigcup \mathcal{F}$ and then $\mathcal{P}(\bigcup \mathcal{F})$.